Understanding public health compliance in emergencies

An investigation of the heterogeneity and pathways of factors that shape our behaviour during public health crises in LMICs

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Introduction

Relevance: Global & Policy Relevance

Pandemic Impact in LMICs

- COVID-19 exposed complex challenges in public health compliance, `especially in low- and middle-income countries`.
- LMICs face unique socio-economic and institutional hurdles not fully captured by HIC-focused research.
- A better understanding of behavioural drivers is critical to design effective, context-specific public health interventions.

Relevance: Enabling better policy making

Predicting policy sucess instead of only ex-post evaluations

- Policymakers require robust, causal evidence to balance health protection with economic stability.
- Identifying the causal pathways in compliance
  - helps avoid unintended policy consequences
  - enables faster and more promising decision making during crises.

> *Is there another pandemic coming? Yes. When? Which pathogen? How severe will it be? No one can say for sure.*

Yonatan Grad, Harvard T.H. Chan School of Public Health

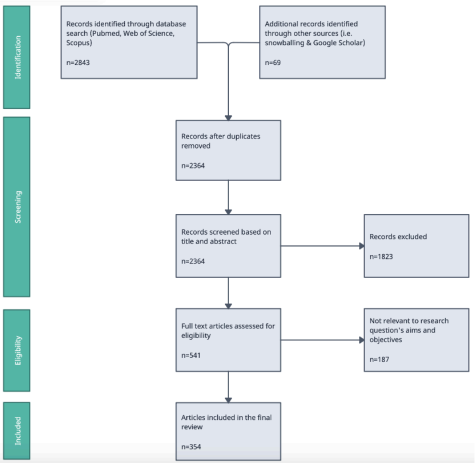
Research Question

How do `dynamic interactions` among diverse behavioural determinants shape public health `compliance` during infectious disease outbreaks in LMICs?

Literature Review

Literature search strategy

- Searched Pubmed, WOS, Scopus
- Added sources through snowballing
- 354 studies reviewed



Peers, Social Cohesion & Norms I

- `Peer Influence and Group Identity`
  - Individuals are more likely to adopt precautions (mask-wearing, distancing) when their immediate social networks and local influencers model these behaviors (Erman and Medeiros 2021; Makridis and Wu 2021).
  - Group identity reinforces compliance when health behaviors are seen as a mark of belonging (e.g. Gelfand et al. 2021).
- `Community Trust & Norms`
  - Tight-knit communities facilitate adherence to health measures (Gelfand et al. 2021).
  - Partisan identities and in-group norms can override other favtors, leading to selective compliance or even counter-normative behavior (Nair, Selvaraj, and Nambudiri 2022).

Peers, Social Cohesion & Norms II

- `Individualism vs. Collectivism`
  - Collectivist societies (e.g., many East Asian countries) achieved higher compliance by emphasizing group welfare, while individualistic cultures (e.g., the US, UK) often resisted restrictions (Gelfand et al. 2021).
- `Hierarchy and Norm Enforcement`
  - High power-distance cultures readily accepted top-down mandates (Alvarez-Galvez et al. 2023)
  - Societies characterized by cultural tightness enforced uniform adherence through strict norms (Neville et al. 2021; Gelfand et al. 2021).
  - Some resisted restrictions citing religious freedom (Filsinger and Freitag 2024).

Health risk perceptions, knowledge & trust

- `Risk perceptions shape compliance intentions`
  - Risk perceptions predict compliance (Bazrafshan et al. 2023; Zucoloto, Meneghini, and Martinez 2022).
- `Knowledge influences risk percetions`
  - Reliable knowledge from public health sources spurred early adoption of measures, whereas misinformation (*infodemic*) undermined compliance (Ferreira Caceres et al. 2022).
  - Trust in institutions (He et al. 2022) & information source consumption (Mohammed et al. 2021) mediate relationship between knowledge and perception.

Trade-offs between life and livelihood

- `Economic Imperatives vs. Health Preservation`
  - Individuals must often choose between immediate economic survival (working despite risks) and protecting their health by complying with mitigation measures.
  - Lower-income groups often faced economic constraints that reduced their ability to comply (Stevens et al. 2021).
  - Studies show that financially constrained individuals are more likely to forgo compliance measures in favor of income security (Miftode et al. 2021; Xiong, Wang, and Zhang 2024).
- If true, in LMICs, where economic vulnerability is pronounced, the trade-off between life (health) and livelihood becomes even more critical

Institutional context, partisanship, and other factors

- Among others, (Keshet 2020) shows effect of institutionl setup, rigidity, and acceptance.
- Strictness of the measure itself also relevant (Guimaraes et al. 2021)
- In highly political contexts, partisanship can override other factors (see e.g. Govind, Garg, and Carter 2024; Filsinger and Freitag 2024).
- Demographic factors sometimes also predictors. E.g. gender (see Zucoloto, Meneghini, and Martinez 2022); Age (Zu et al. 2021)
  - But highly correlated with other factors (like risk perception, attitudes, etc.)

Gaps I

Peer group effects vs self-selection

- Unclear whether group/ network effects are driven by self-selection into groups
- Some descriptive insights from self-reported reasons for behaviour
- Evidence from influenza suggest *pure* peer effects in vaccination behaviour (Miyazato, Ibuka, and Itaya 2024)

Gaps II

Is there a health vs income trade-off?

- Trade-off widely assumed but never causally confirmed and quantfied
  - Influence of economic status & health risk perception confirmed seperately
  - Most studies on role of economic status done in HICs
  - Not yet modelled together and isolated from other factors that might *override* trade-off
  - Some WTP insights for vaccines (e.g. Catma and Varol 2021), but no quantification of impact of health risk perceptions **relative to economic vulnerability** and vice versa on NPI behaviour

Gaps III

Dynamic interactions

- Factors mostly modelled in isolation
- Crucial interactions sometimes assumed but never causally traced
  - Eg interaction of risk perceptions vs economic vulnerability
- Limited application of hollistic frameworks in HICs
  - EG Health belief model in USA (Badr et al. 2021), Theory of planned behaviour in France (Wollast et al. 2021)
  - One study among students in Vietnam (Bui et al. 2023)
  - Exclusively single country, single time-period studies

Aims & research frameworks

Chapter	Research question
1	What are the mechanisms and interactions through which economic, socio-cultural, institutional, and psychological determinants shape individual compliance with public health measures during pandemics in LMICs?
2	What is the relative importance of economic constraints and health risk perceptions on compliance with public health measures?
3	What is the effect of membership in non self-selected peer groups on vaccination intentions?

Chapter I: Informing a theory of pandemic behaviour

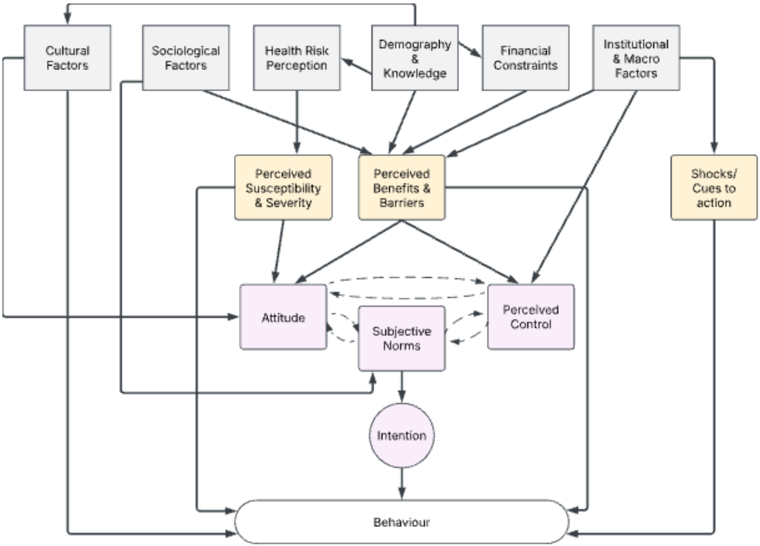
RQ: What are the mechanisms and interactions through which economic, socio-cultural, institutional, and psychological determinants shape individual compliance with public health measures during pandemics in LMICs?

Background & Gaps I

Different factors interact and shape behaviour **together**

- Some studies use theoretical frameworks (Health Belief Model `OR` Theory of Planned Behaviour) to analyse survey data
- Only (Bui et al. 2023) in LMIC (Looking at students in Vietnam)
- No paper integrating/ extending these in Covid 19 in LMIC
- No study comparing different geographical contexts
- No comparison of NPI and vaccines

Background & Gaps II



Data

MIT C19 preventative health survey

- Conducted via online surveys distributed to Facebook users globally
- Over 100k observations across 78 countries
- Weighted to account for probability of being FB user
- Contains data on all aforementioned factors
- Repeated cross sectional survey
- Requesred & granted. Currently correspondence between LSE Datalibrary and MIT.

Methods I

Approach

- Structural equation modelling to uncover latent and observed pathways
  - Explicitly models causal pathways and feedback loops
  - Estimated using system modelling like maximum likelihood, Generalized least squares, Bayesian estimation
- Examples of application are
  - Newton et al. (2024) modelling Health outcomes as a function of social determinagts of health in UK.
  - Li et al. (2020) modelling dementia care ability as a function of health literacy and social support in China

Methods II

Approach

- Structural equation modelling to uncover latent and observed pathways
  - Allows simultaneous estimation of direct and indirect effects in predicting preventive behavior
- Direct, indirect, and total effects analysed for all models
- Conduct at different waves for robustness checks and comparison
  - Evaluation of goodness of fit between HBM, TPB, extended and integrated version using AIC/BIC
  - Evaluation of overfitting via multi-round application of estimated parameters and assesment of goodnes of fit variation

Expected Contributions

Emprical contribution

- First large scale SEM across countries and waves
- Qunatification of uncertainity and relative importance of factors

Theoretical Contribution

- Informing more comprehensive framework of pandemic behaviour
- Introducing nuance around different intervention types (NPIs vs vaccine)

Policy relevance

- Informs more efficient data collections during early phases of emergencies
- Informs ressource allocation efficiency

Chapter II: Qunatifying the health-income trade-off assumption

RQ: What is the relative importance of economic constraints and health risk perceptions on compliance with public health measures?

Background & Gaps

Widely assumed trade-off between income an health that `individuals` face

- Rationale behind cash transfer programmes
- Supported by economic theory (Grossman 1972); Prospect Theory
- Empirical eveidence shows
  - Effects of income (e.g. Stevens et al. 2021)
  - Effects of risk perceptions (e.g. Bazrafshan et al. 2023)
- Income and risk perception never modelled together to confirm `trade-off` in causal way

Hypotheses

H1(2): Economic vulnerability (health risk perceptions) predict compliance.

H3: Individuals face a trade-off between preserving their health and their economic security.

Data

World Bank Malawi high frequency phone survey between 2020 and 2024

- Subset: 05/20-07/21
  - Only the first 13 rounds contain information about non-pharmaceutical mitigation behaviour
- Nationally representative sample of households based on the Malawi National Integrated Household Panel Survey from 2019
- 1729 participants at baseline & 8% attrition at R13
- Further weighting based on census data

Methods: Double Debiased Machine Learning I

- Traditional econometric approaches
  - Identification of pure effect requires isolation from confounding
    - Including limited confounding information -> endogeneity concerns
    - Including a lot of controls -> modelling challenges due to complex functional forms (eg colinearity, non-linearity, etc.) (Belloni et al. 2017)
- Machine Learning
  - Can handle complex functional forms and data patterns (eg many non-linear controls, colinearity) for predicting outcomes.
  - Delineating different factor's distinct influence challenging.
  - Estimates difficult to interpret due to (often) non-tracable data trahsformations.

Methods: Double Debiased Machine Learning II

Combining the benefits of both approaches

- Proposed by Chernozhukov et al. (2018)
- Predict behaviour & treatment separately given vector of covariates using ML methods
- Regress residuals of predicted outcome (i.e. the variation in outcome unexplained by covariates) on residuals of predicted treatment (ie the variation in treatment not explained by covariates)
  - Estimate = Causal effect of treatment on outcome
  - Based orthogonalization proposed by Neyman and Scott (1948)

Expected Contributions

Emprical contribution

- First causal investigation into health/income trade-off in context of protective measures during a pandemic
- Qunatification of trade-off and comparison over time

Theoretical Contribution

- Empirical test of widely-assumed trade-off assumption

Policy relevance

- Informs more efficient data collections during early phases of emergencies
- Informs ressource allocation efficiency

Chapter: The role of peer effects on vaccination intentions

RQ: What is the effect of membership in non self-selected peer groups on vaccination intentions?

Background & Gaps

Methods

Data

Expected Contributions

Emprical contribution

Theoretical Contribution

Policy relevance

Thank you!

Questions & Feedback

Annex: Literature search strategy II: Search string

```
TITLE-ABS-KEY(
  (pandemic* OR epidemic* OR outbreak* OR "public health cr
  AND
  (behavio*r OR compliance OR adherence OR "preventiv* meas
    OR lockdown* OR "stay at home" OR "social distan*" OR c
    OR "vaccine acceptance" OR "vaccination intent*" OR "te
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  AND
  (economic OR "socio-cultural" OR social OR institutional
    OR "peer effect*" OR "media influence*" OR politic* OR
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Annex I: Chapter I

Annex: Structural Equation Modelling II (Skip)

Confirmatory Factor Analysis (CFA)

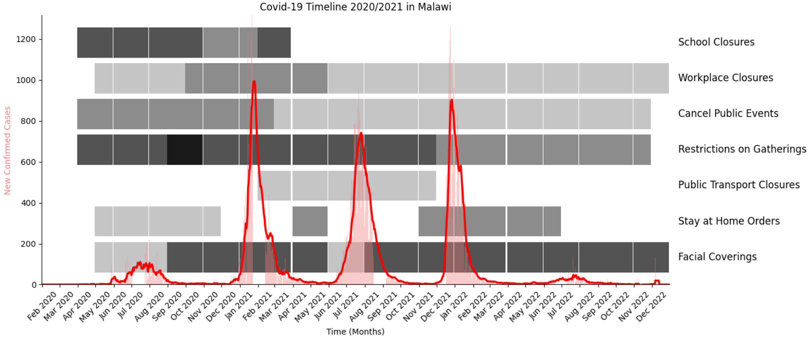
- To validate latent constructs
- To operationalise constructs from survey items

Comparative Model Evaluation

- Predictive power assessment (variance explained in preventative behavior)
- Multi-group SEM for cross-country/SES/gender comparisons
- Akaike Information Criterion (AIC) & Bayesian Information Criterion (BIC) for model comparison

Annex II Chapter 2

Annex: Chapter II BG: Malawi during waves 1 & 2 of C19



Outcome	Indicators	Investigation at	Hypothesis tests
Social distancing	Gathering attendance; non-essential travels; Reduced Shopping	Multiple rounds	H1, H2, H3
Hygiene & Mask	Hand Washing; Mask Wearing	Single Round	H2
Willingness to test*	Willingness to test	Single Round	H1
Vaccine Acceptance*	Willingness to vaccinate	Single round	H1

*Vaccines and tests are provided for free but involve significant transaction and opportunity costs. Reddy, K.P., Fitzmaurice, K.P., Scott, J.A. et al. Clinical outcomes and cost-effectiveness of COVID-19 vaccination in South Africa. Nat Commun 12, 6238 (2021)

Annex: Chapter II How: Treatments I

Economic Vulnerability Indicator

- Constructed from factors such as:
  - Current income
  - Self-reported economic stability perception
  - Economic risk perception due to the pandemic
  - Main income source (formal vs. informal economy)
  - Consumption quantile (based on monthly spending)
- Operationalization methods under consideration:
  - Inverse Probability Weighting (IPW) by Anderson
  - Established economic vulnerability measures (e.g., Calvo & Dercon, 2005)
  - Data-driven approach (PCA or single factor analysis)

Annex: Chapter II How: Treatments II

Health Risk Perception Indicator

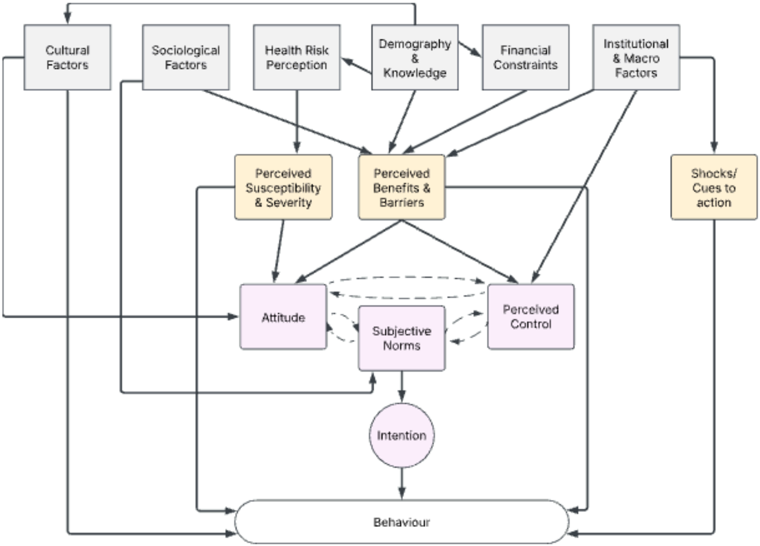
- Self-reported health risk perception
- Infection history and self-reported comorbidities
- Recent Covid-19 symptoms

Trade-off as interaction term

Annex: Chapter II How: Controlling for ...

- Demographics (Age, gender, education, urban/rural, etc.)
- Cultural affiliations and beliefs (religion, parties, norms, etc.)
- HH characteristics (size, adult equiv., etc.)
- Exposure to C19
- Trust in institutions & conspiracy beliefs
- Mental health
- Macro context (policies, economic performance, etc.)
- And some more (self-efficacy, C19 communication exposure, news consumption, etc.)

Annex: Chapter II How: Challenges of traditional econometric approaches I



Annex: Chapter II How: Challenges of traditional econometric approaches II

Annex: Chapter II How: Challenges of machine learning

Strengths

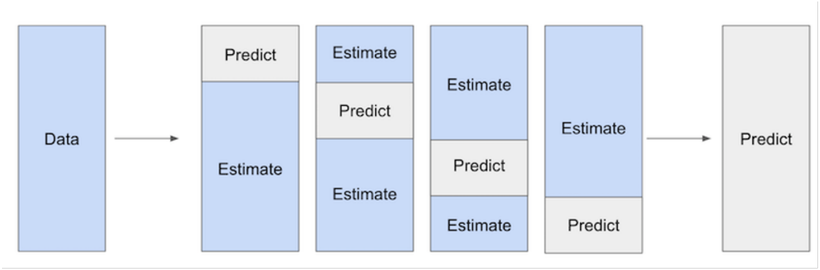
- Very flexible re functional forms
- Can handle large amounts of covariates

Limitations

- Overfitting concerns
- Causal pathway interpretation challenging

Annex: Chapter II How: Double Debiased Machine Learning II

Cross-fitting to reduce overfitting and regularization bias



Annex III Chapter 3

Alvarez-Galvez, J, A Anastasiou, D Lamnisos, M Constantinou, C Nicolaou, S Papacostas, VS Vasiliou, et al. 2023. "The Impact of Government Actions and Risk Perception on the Promotion of Self-Protective Behaviors During the COVID-19 Pandemic." *PLOS ONE* 18 (4). https://doi.org/10.1371/journal.pone.0284433.

Badr, Hoda, Xiaotao Zhang, Abiodun Oluyomi, LeChauncy D. Woodard, Omolola E. Adepoju, Syed Ahsan Raza, and Christopher I. Amos. 2021. "Overcoming COVID-19 Vaccine Hesitancy: Insights from an Online Population-Based Survey in the United States." *Vaccines* 9 (10): 1100. https://doi.org/10.3390/vaccines9101100.

Bazrafshan, A, A Sadeghi, MS Bazrafshan, H Mirzaie, M Shafiee, J Geerts, and H Sharifi. 2023. "Health Risk Communication and Infodemic Management in Iran: Development and Validation of a Conceptual Framework." *BMJ OPEN* 13 (7). https://doi.org/10.1136/bmjopen-2023-072326.

Belloni, A., V. Chernozhukov, I. Fern ndez-Val, and C. Hansen.